VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITI-HT-0050

Dermasense: a laser-written strip designed to read a dressing's own 'change line'

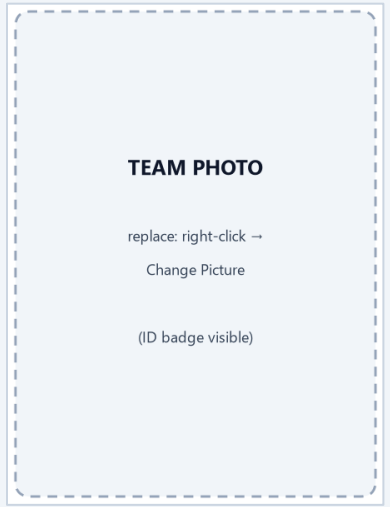
Soham Prabhakar Kadu · Bhakti Ajay Kadam · Sanjivani University

Health Tech: Emergency Response & Remote Monitoring

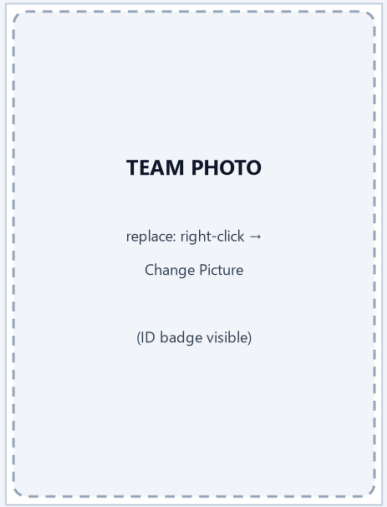
Evidence key: • verified • designed, lab test pending • future scope

Project Overview & Team

VK26-IITI-HT-0050	Health Tech	Sanjivani University	Stage 2
TEAM / APPLICATION ID	EMERGENCY RESPONSE & REMOTE MONITORING	INSTITUTION	DESIGN SPECIFICATION · OCT 2026



Soham Prabhakar Kadu
Role: [team to add]
Sanjivani University

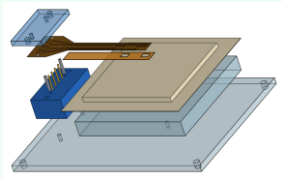


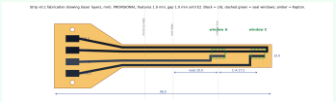
Bhakti Ajay Kadam
Role: [team to add]
Sanjivani University

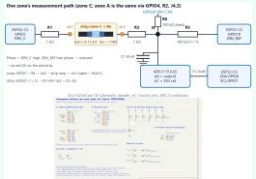
Team Dermasense (photos taken by the team).

What we built in Stage 2 (14–19 Oct 2026)

- **CAD:** 9 SolidWorks parts, 1 assembly, 7 A3 drawings; 99 automated checks pass.
- **Circuit:** KiCad reader schematic; 0 rule-check errors.
- **Firmware:** ESP32-C3 reader with Bluetooth output; 23 + 16 software tests pass.
- **Sensor strip:** laser files for the strip and 2 test coupons; 35/35 file checks.
- **Experiment:** E2 laser test, 72 samples fixed in advance.
- **Not yet:** nothing lased, wired or tested on a dressing.


SolidWorks model


Laser file (DXF)


Reader circuit (KiCad)

Core Innovation: two laser-written zones, at the pad centre and on the maker's change line, let a clip-on reader tell a carer OK, CHANGE or CHECK without lifting the dressing.

Problem Context

PUBLISHED EVIDENCE · CITED

44.9 %	≈ 2 cm	1 point	Uneven
OF 588 CHANGES MADE AT OPTIMAL MOISTURE[1]	MAKER'S 'CHANGE NOW' LINE FROM PAD EDGE[2]	WHERE TODAY'S MOISTURE SENSOR READS[1]	IN-VITRO MOISTURE MAP, 8 POINTS[3]

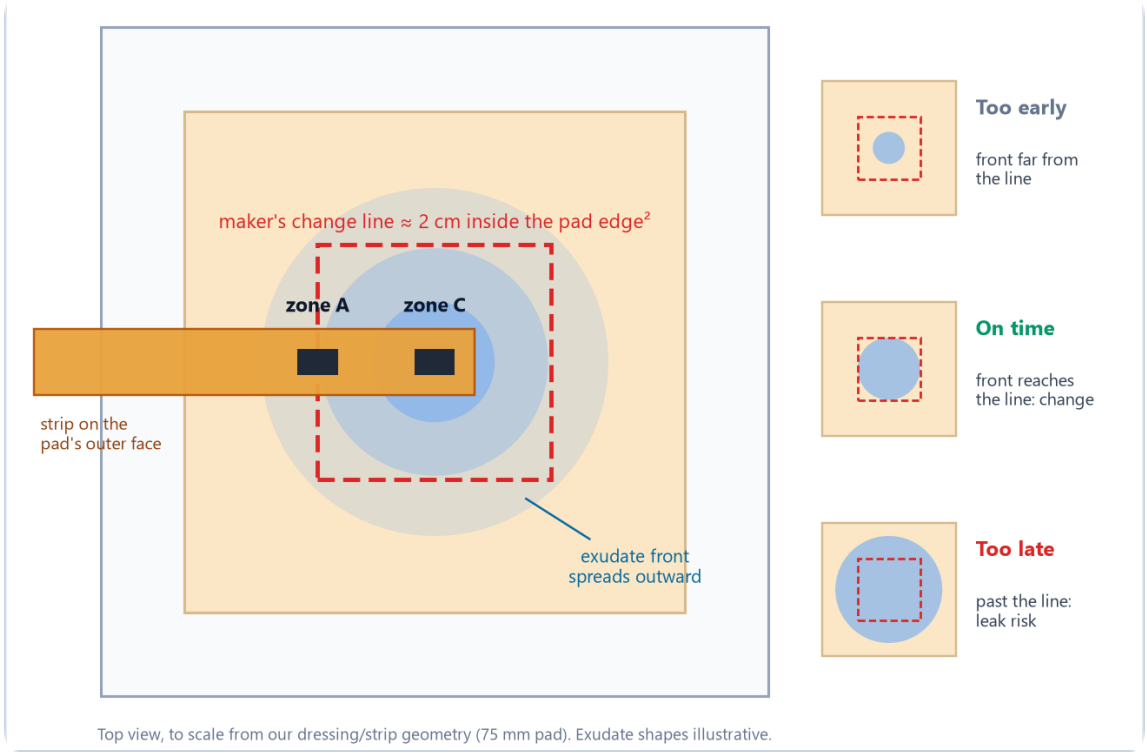


Figure: the change line is invisible under a cover (to scale, our design).

Target Beneficiary & Problem Statement

- **Who:** nurses and home carers of chronic wounds under foam dressings.
- **Pain:** the maker's 'change now' sign is visual[14], and the dressing is covered.
- **Today:** change by the clock, by lifting the cover, or after a leak.
- **Gap:** in our search, no low-cost sensor reads that change line[45].

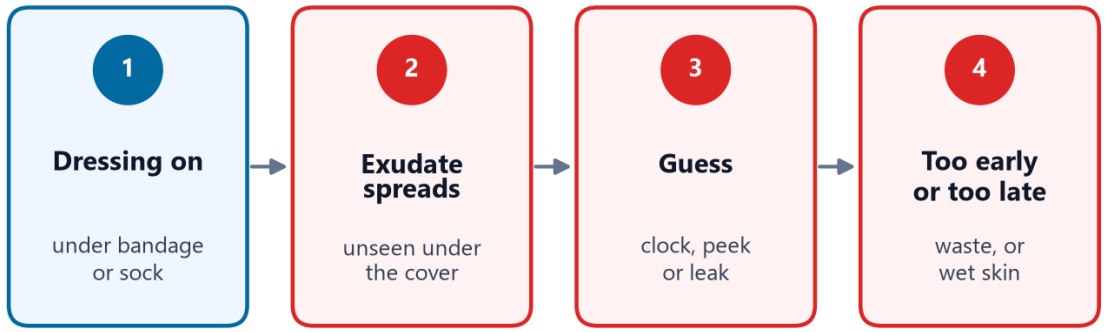
Problem: carers cannot see when a covered dressing is really full.

User Journey: Before State

CURRENT STATE · CITED

44.9 %	≈ 18 min	₹240–510	0.1–1.3 g/cm²·day
CHANGES WHILE MOISTURE WAS STILL OPTIMAL[1]	A DRESSING VISIT, PATIENT-REPORTED (UK)[13]	ONE 10 × 10 CM FOAM DRESSING, RETAIL[15]	REPORTED EXUDATE RATES, 2 METHODS[6]

Today: checked by schedule or by looking



44.9 % of changes came while moisture was still optimal[1]

Cited numbers only (References slide)

Figure: today's pain-point map (cited numbers only).

Current Pain Point Mapping

- **Time:** every check means unwrap, look, re-secure: about 18 min a visit[13].
- **Waste:** 44.9 % of changes came while moisture was still optimal[1]; each discards a dressing[15].
- **Risk:** a late change can leak: wet skin, soiled clothing.
- **Cause:** reported exudate rates span over 10×[6]; a clock cannot follow.

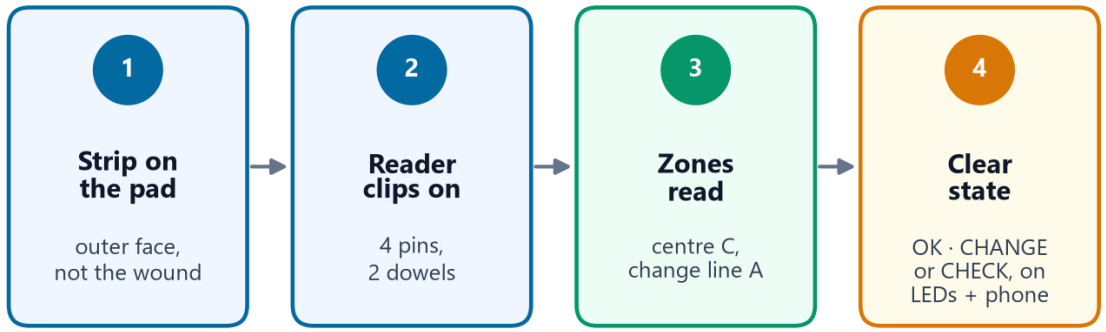
Quantified Inefficiency: published figures only (a UK patient survey, Indian retail prices); we will time real checks with nurses.

User Journey: After State (Intended)

INTENDED · NOT CLINICALLY VALIDATED

up to 44.9 %	≤ ₹100	±10 %	0
EARLY CHANGES A RELIABLE ALERT COULD AVOID[1]	TARGET STRIP COST (ESTIMATE)	TARGET ALERT ACCURACY (BENCH, E4)	CONTACT WITH THE WOUND BED (BY DESIGN)

Intended: the strip reads the change line



Target: alert within ±10 % of the visible front (bench test E4)

INTENDED WORKFLOW · NOT CLINICALLY VALIDATED

Figure: the intended workflow; bench-testable, not clinically validated.

Future Solution Mapping & Quantified Efficiency

- **Same routine:** strip on the pad, reader clips on.
- **Clear signal:** CHANGE when the line is wet; CHECK if it wets first.
- **Remote:** designed to broadcast the state to a nearby phone over Bluetooth (hardware test pending).
- **Payback:** a ≤ ₹100 strip pays for itself if 1 in 2–5 changes is avoided[1][5] (strip only; the reader is reused).

Design Principle: only film and carbon are disposable; the strip never touches the wound.

System Architecture & Specs

DESIGNED · LAB TEST PENDING

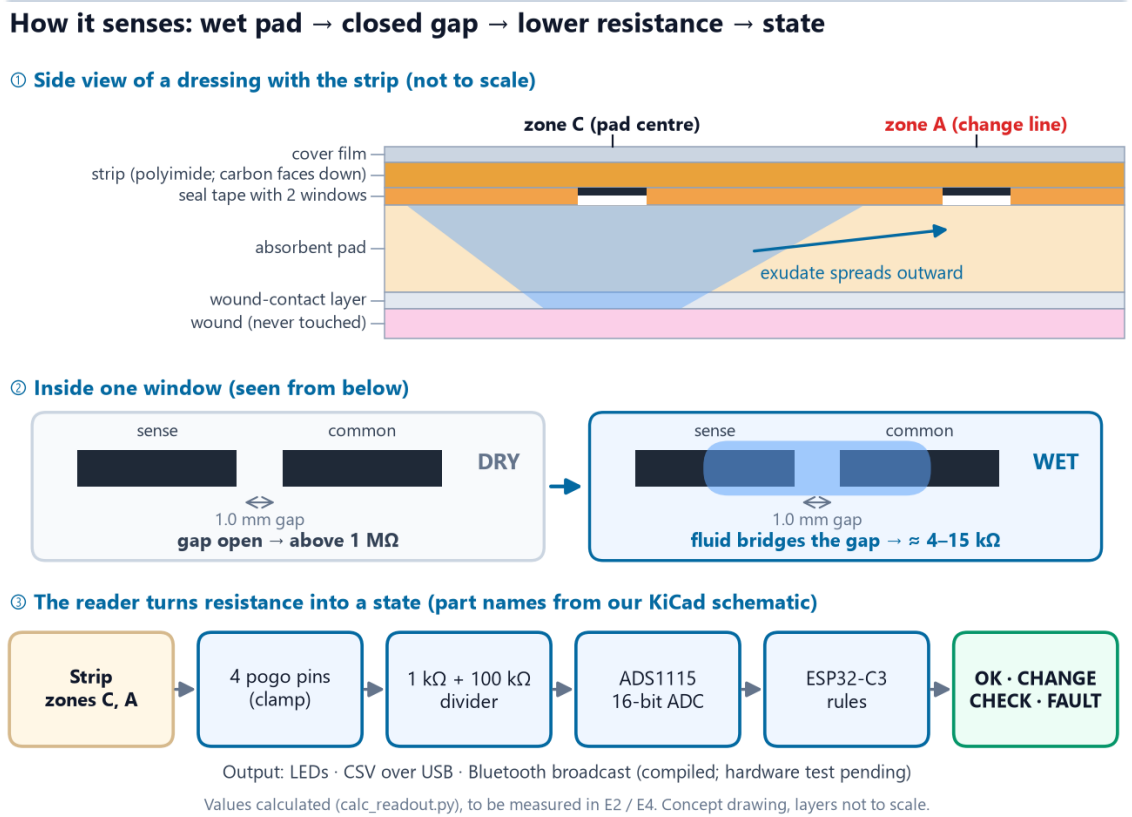


Figure: how a wet pad becomes a state (concept drawing; values calculated, to be measured).

Target Specifications & Hardware

Parameter	Design value	Basis
Sensing zones	2: centre C, change line A	decision D-012
Electrodes	1.0 mm lines and gaps	E2 confirms
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans	calc: wet 4–15 kΩ
Drive	bipolar, ≤ 33 µA, no net DC	calc · bench T8
ADC / MCU	ADS1115 16-bit (I²C) · ESP32-C3	datasheets
Self-test	100 kΩ check ±5 % + strip loop	else FAULT
Contacts	4 pogo pins + 2 dowels	CAD checked
Output	LEDs · USB log · BLE broadcast	BLE compiled only

Architecture Rationale: only film and carbon are disposable; every electronic part is reused.

System Architecture & Specs: Firmware & Data Flow

VERIFIED IN SOFTWARE · HARDWARE PENDING

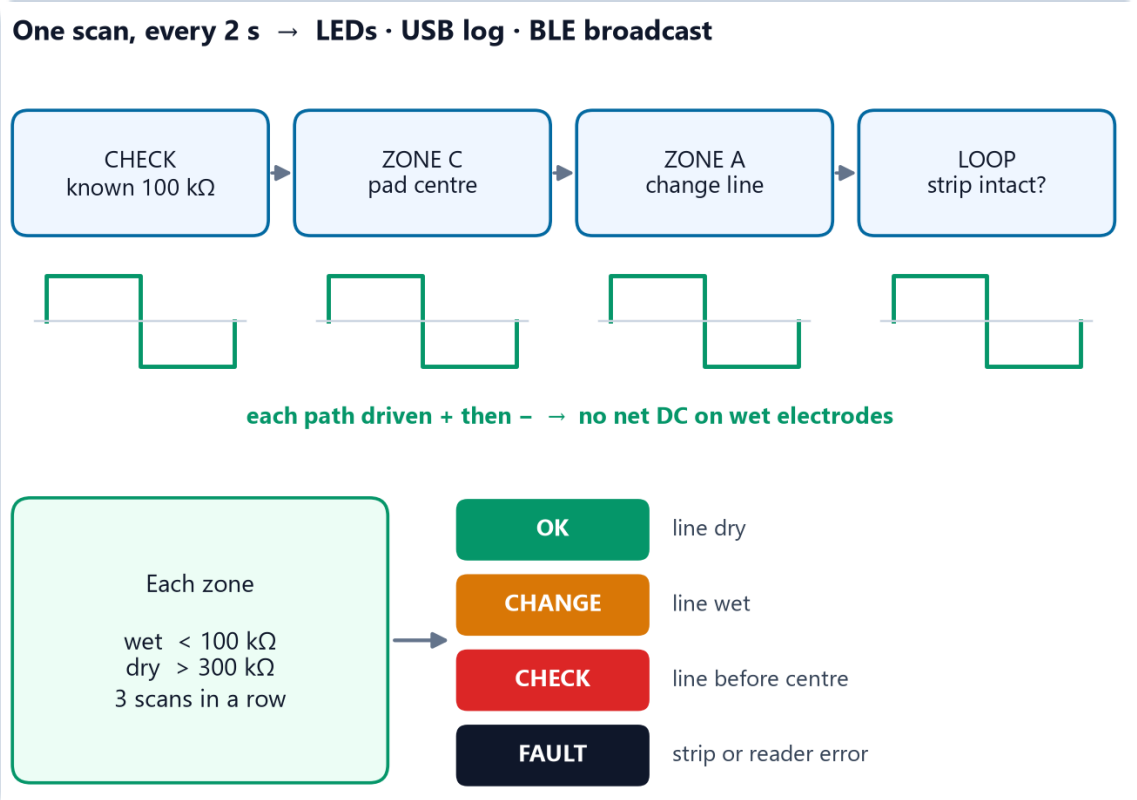


Figure: one reader scan, from measurement to state.

Firmware Logic & Data Flow

- **Measure:** each zone driven both ways, so no net DC.
- **Read:** 16-bit ADC, scaled to the 3V3 rail.
- **Decide:** wet < 100 kΩ, dry > 300 kΩ, 3 scans in a row.
- **Self-check:** a known 100 kΩ every scan, else FAULT.
- **Output:** LEDs; a CSV line over USB; a Bluetooth broadcast any scanner app can read (no custom app yet).
- **Why remote:** a carer nearby can see the state without unwrapping or plugging in (to be bench-tested).

Software vs Hardware: 23 compile-time and 16 PC tests pass; not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT

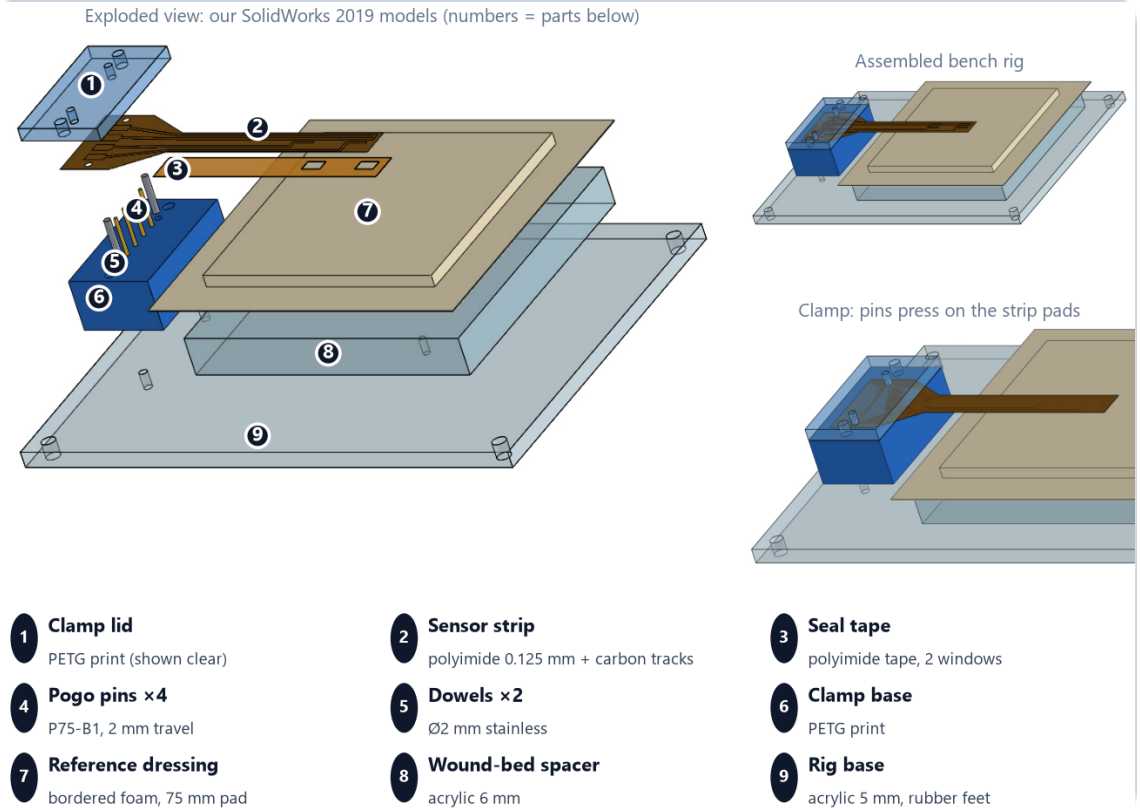


Figure: our SolidWorks 2019 models (9 parts, 1 assembly, 7 A3 drawings, STEP); not yet built.

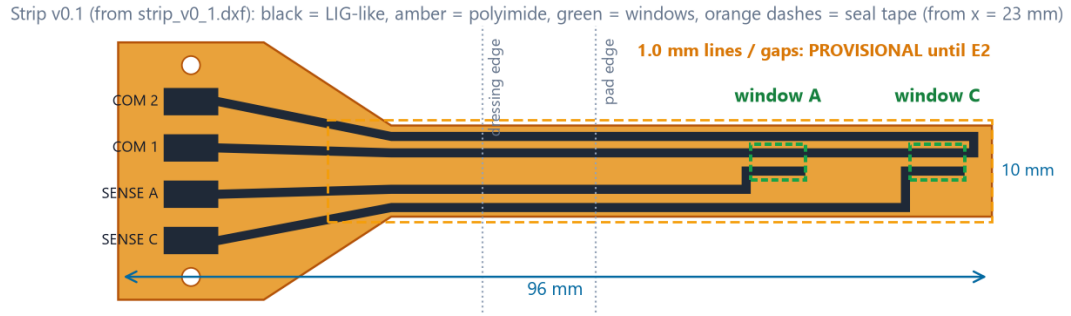
A clip-on clamp: 4 spring pins, 2 dowels

Check	Result	Basis
Materials	polyimide strip · PETG clamp · acrylic rig	printable, laser-cut
Pin force	0.72 N each · 2.9 N total	calc
Lid bending	11 µm vs 1.48 mm pin travel	calc
Alignment	0.49 mm used of ±1.35 mm	calc · 2.8× margin
Strip bending	0.62 % strain at 10 mm radius	calc · yield ≈ 3 %
CAD checks	99 pass · 0 fail · 5 wait for E2	verify_cad.py
FEA	not run: loads are under 3 N	hand calcs above

Design Iteration: a clash check found the seal tape trapped under the clamp (9.0 mm³); moved to x = 23 mm, now 0 clash (decision D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING



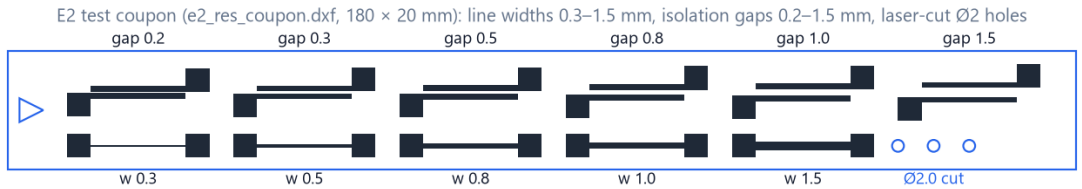


Figure: our laser files, drawn from the DXFs. Not yet lased.

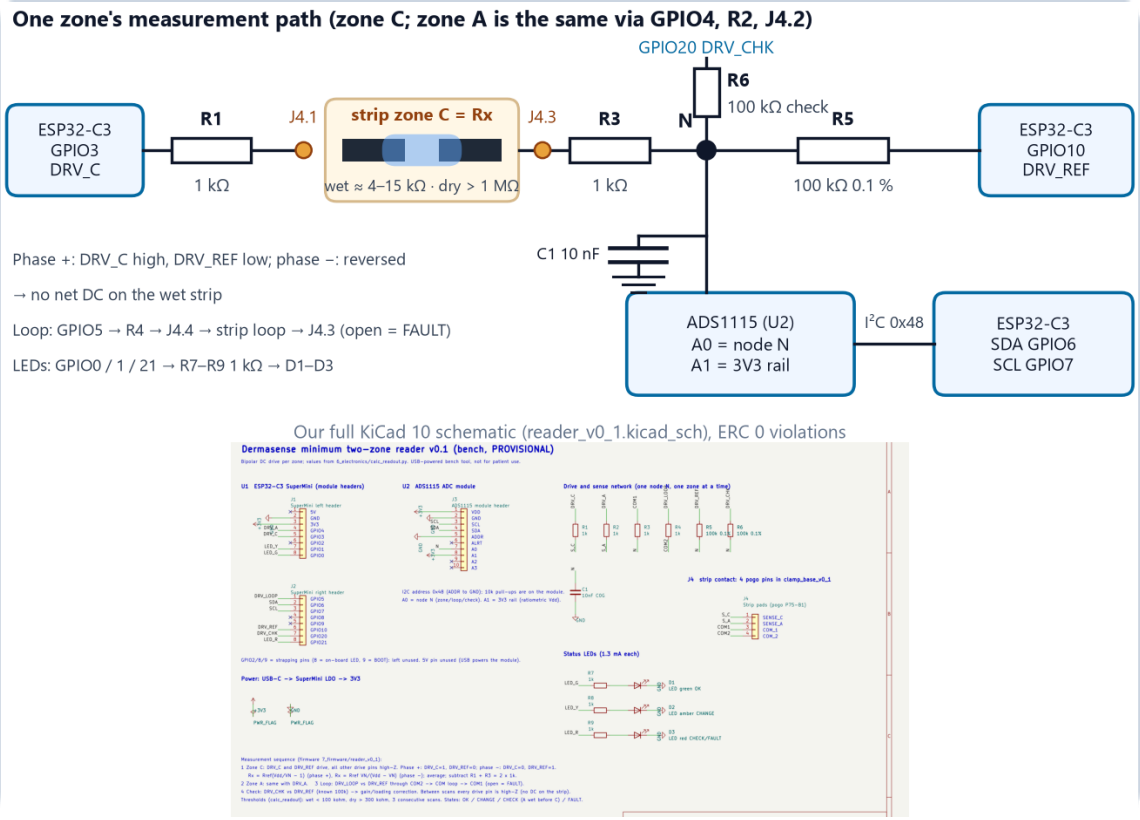
Fabrication & Process Validation (E2)

- **Material:** laser-carbonised, LIG-like layer on polyimide[7].
- **Reference:** setting '12', most likely % PWM, not watts.
- **E2 plan:** 7 coupons, 2 days, 72 pre-registered samples.
- **Pass rules:** CV ≤ 20 %, gaps > 3 MΩ, fixed beforehand.
- **Sealed:** leads under tape; LIG made in air wets easily[9].

Laser Files Checked: 35/35 file checks pass (files only; nothing lased yet). LaserCAD import is checked at the machine.

Electronics Schematics & BOM

ERC VERIFIED · NOT YET WIRED



Our full KiCad 10 schematic (reader_v0_1.kicad_sch), ERC 0 violations

Figure: zone-C path drawn from our netlist, above our real KiCad 10 sheet.

Pin & Net Map (from the KiCad netlist)

Net	Pin	Function
DRV_C · DRV_A	GPIO3 · GPIO4	drive zone C · A via 1 kΩ
DRV_LOOP	GPIO5	strip loop (broken strip)
DRV_REF	GPIO10	100 kΩ 0.1 % reference
DRV_CHK	GPIO20	100 kΩ self-check
SDA · SCL	GPIO6 · GPIO7	I²C to ADS1115 (0x48)
N · 3V3	ADS1115 A0 · A1	node · rail (ratiometric)
LED G · Y · R	GPIO0 · 1 · 21	OK · CHANGE · CHECK/FAULT
Strip pads	J4.1 – J4.4	S_C · S_A · COM1 · COM2
Power	USB-C → 3V3	LDO on the SuperMini

Verification: ERC 0 violations; netlist matches firmware and strip pads (0 mismatches).

Electronics Schematics & BOM: Bill of Materials

3 PRICES QUOTED · NOTHING ORDERED

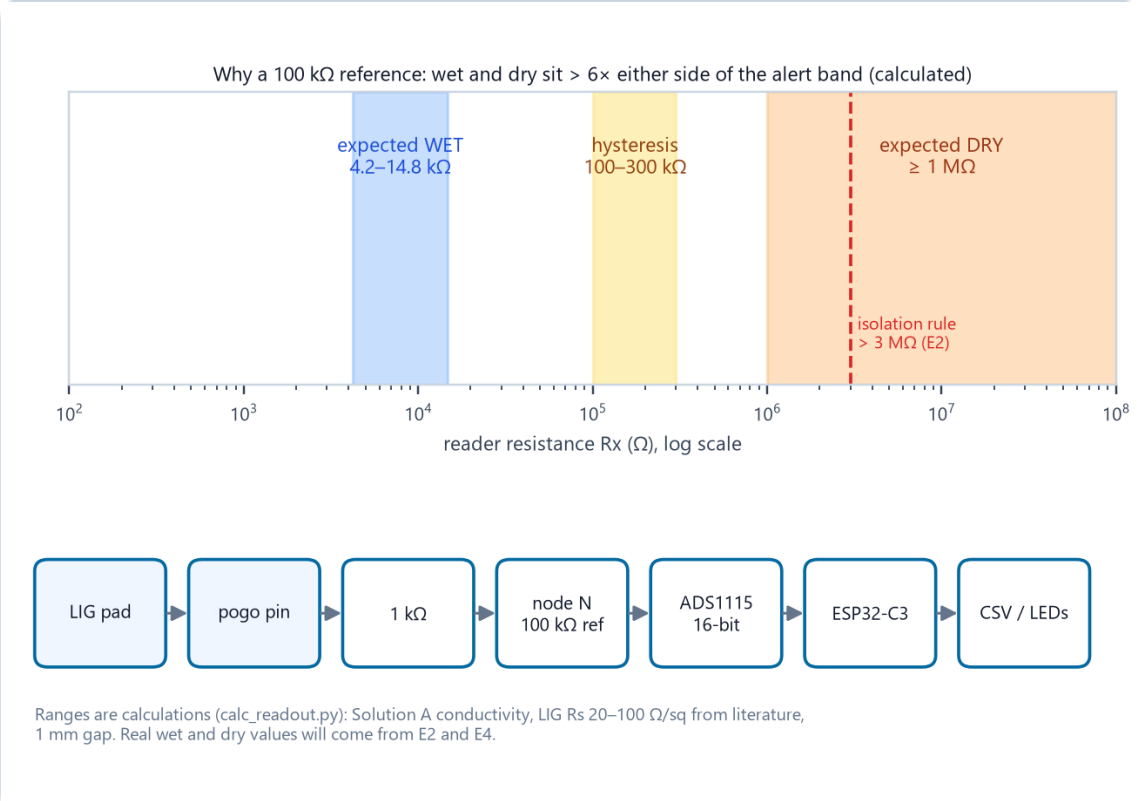


Figure: why a 100 kΩ reference (calculated ranges).

Prototype Bill of Materials

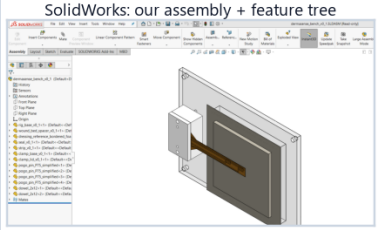
Item	Qty	Source	₹
ESP32-C3 SuperMini (PRO4607)	1	Probots · quoted	242
ADS1115 module (PRO789)	1	Probots · quoted	125
P75-B1 pogo pins (05880)	100	Hubtronics · quoted	354
Resistors, C0G cap, LEDs, board	set	local / online	158–421
Clamp + bench rig	set	lab / local	100–1,240
Strip materials	set	lab / online	800–3,500
Test consumables	set	local	480–1,060
Total			**≈ 2,259–6,942**

Procurement Status: reader parts and pins quoted 19 Oct, in stock in India; the rest are estimates. The reader is reused; only the strip is consumed.

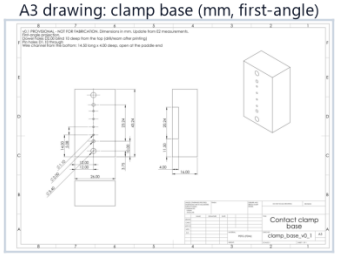
Maker Journey

REAL FILES · RE-CHECKABLE · NO STAGED PHOTOS

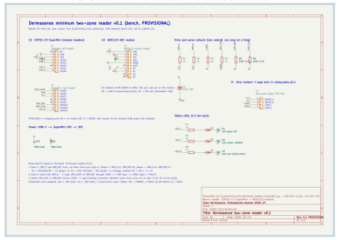
SolidWorks: our assembly + feature tree



A3 drawing: clamp base (mm, first-angle)



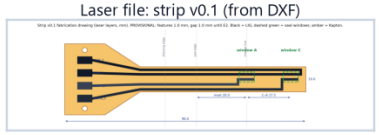
KiCad 10: reader schematic



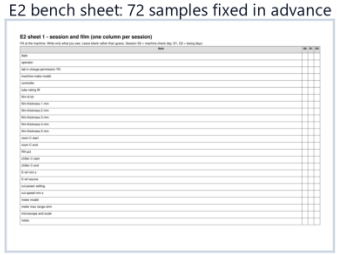
Firmware: state rule + compile-time test

```
// C = pad centre, A = maker's change line
constexpr State classify(Wet c, Wet a,
                         bool fault) {
  return fault ? State::Fault
    : (a == Wet::WET ? (c == Wet::WET
         ? State::Change : State::Check)
       : State::Ok);
}
static_assert(classify(Wet::WET, Wet::WET,
   false) == State::Change,
   "front reached the change line");
```

Laser file: strip v0.1 (from DXF)



E2 bench sheet: 72 samples fixed in advance



Checks re-run on 19 Oct 2026 (results as printed by each tool)

verify_cad.py (SolidWorks)	{'PASS': 99, 'FAIL': 0, 'BLOCKED': 5}
kicad-cli sch erc	Found 0 violations
verify_laser_files.py	35 PASS, 0 FAIL
arduino-cli compile (esp32c3)	Sketch uses 650349 bytes (49%)
pytest (firmware tools)	16 passed
pytest (E2 analysis, synthetic data)	3 passed

Figure: our own files and tool output (checks re-run 19 Oct 2026).

Authenticity: real, dated, re-checkable work

LAB PHOTO

team at work, ID badges visible

replace: right-click → Change Picture

LAB PHOTO

team at work, ID badges visible

replace: right-click → Change Picture

Team at work, institutional ID badges visible (photos by the team).

- **Real files:** .SLDPRT .SLDASM .SLDDRW .STEP .kicad_sch .ino .dxf, dated in 15 git commits since 14 Oct.
- **Re-checkable:** one command re-runs each check shown on the left.
- **How we work:** scripts written with AI assistance (Claude), directed by us, build the CAD, schematic and firmware; the physical build and tests are done by us, starting with E2.
- **Photos:** never staged or generated.

Raw Repository Linkage: [team to insert the Drive link before upload]: all native files and a dated work history.

Maker Journey: Engineering Iterations

DOCUMENTED DECISIONS

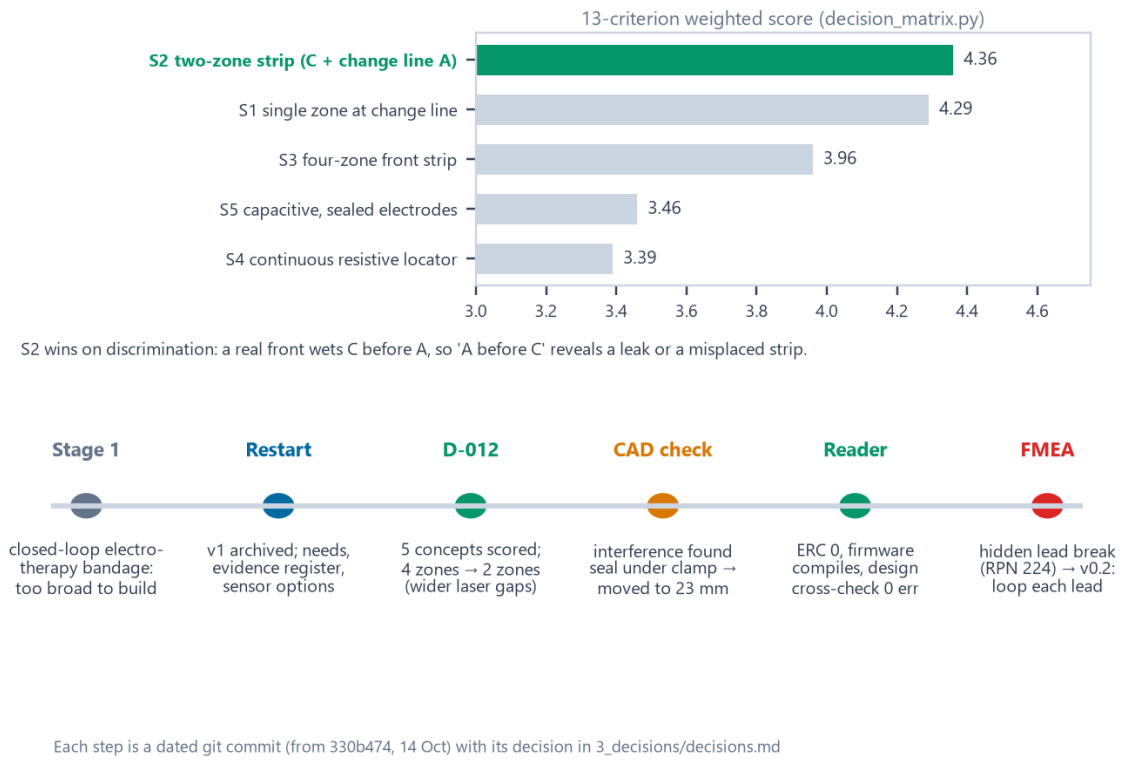


Figure: concept scores and the iteration path (each a dated commit).

Stage 1 to Stage 2 Iterations

- **Pivot:** therapy bandage (already shown in labs[5]) → one job: is it full?
- **Choice:** two-zone strip (4.36) beat one zone (4.29): it spots leaks.
- **Kept:** Stage 1's Bluetooth telemetry, now a state broadcast (compiled; hardware test pending).
- **Corrected:** Stage 1 called Kapton 'medical-grade'; DuPont cautions otherwise[11].
- **Redesign:** FMEA risk → v0.2 loops each sense lead.

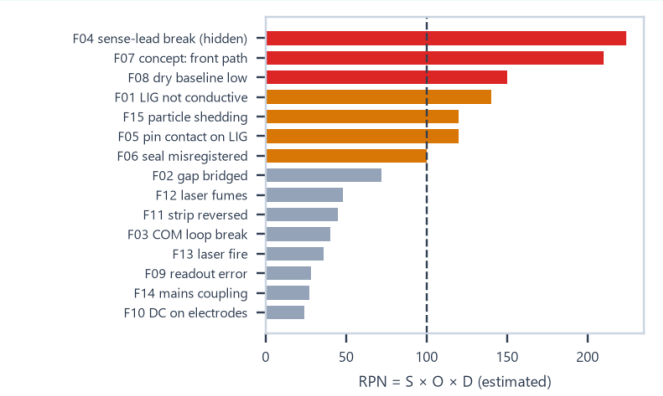
Prior Art: diaper strips[4] and visual change indicators[14] exist; our novelty is reading the change line electrically, with an order check.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

Design FMEA: top 5 of 15 failure modes (ratings estimated before tests)

Failure mode	Effect	RPN	Fix / test
F04 cracked sense lead	zone reads 'dry' for ever	224	v0.2 loops each lead · T-MECH-01
F07 front path differs	CHANGE late or never	210	E4 on real foam dressings
F08 humid dry baseline	false 'wet'	150	measure baselines (E4)
F01 weak carbon layer	zones always 'dry'	140	E2 + per-strip QC record
F05 pin contact on carbon	readings drift high	120	rounded pins · T-CON-01



Fail-safes

- **Bench only:** USB reader, never on a person.
- **No net DC:** bipolar drive, idle pins off; bench test T8 checks < 5 mV.
- **Self-check:** every scan, else FAULT.
- **No wound contact:** strip on the pad's outer face, leads sealed.
- **Laser work:** signed safety checklist every session.

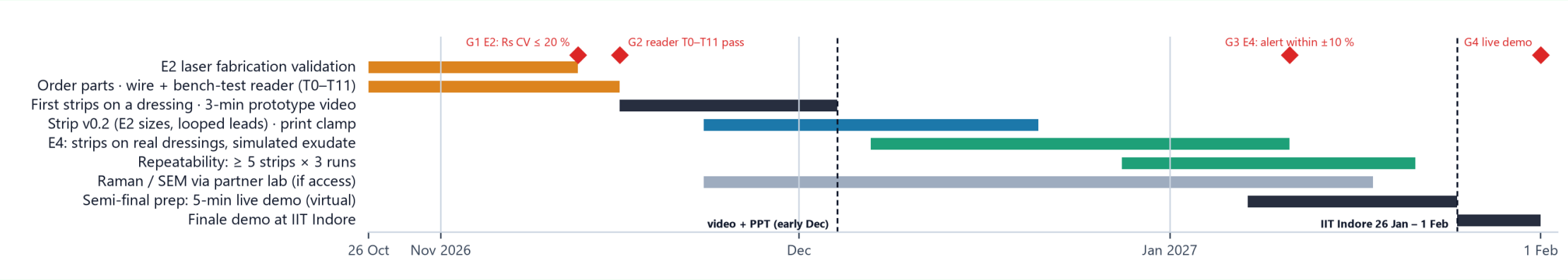
Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

FUTURE SCOPE · GATED PLAN

4-Month Plan on the programme's own dates: each gate must pass first



Build & Bench (Months 1 - 2)

- **Nov:** E2 laser tests (G1); build and bench-test the reader (G2).
- **Early Dec:** first strips on a dressing; 3-min prototype video.

Validate & Demo (Months 3 - 4)

- **Dec:** strip v0.2 + printed clamp; E4 wet tests (G3).
- **Jan:** repeatability; virtual semi-final with a live demo.
- **Stage 3:** working strip + reader: OK → CHANGE live on a dressing, raw data logged.

Technical Challenges & Risks

FOUND IN OUR OWN VERIFICATION

Geometry: Seal Under Clamp

Found: seal tape overlapped the clamp (9.0 mm³).

Impact: would lift the strip off the pins.

How: automated clash check.

Circuit: Hidden Lead Break

Found: a cracked sense lead reads 'dry'.

Impact: CHANGE never shown, no alarm.

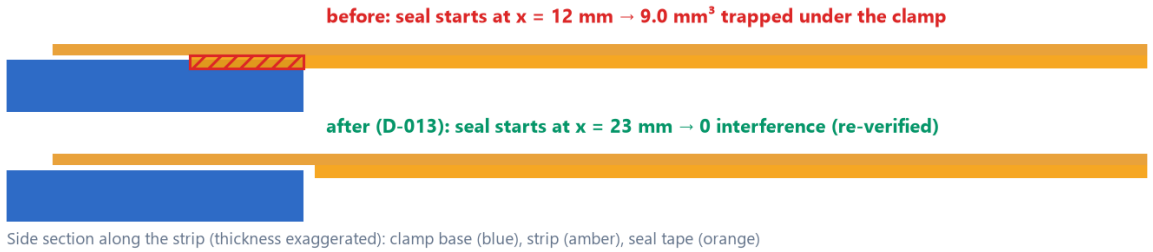
How: design FMEA (RPN 224).

Process: Unknown Laser Recipe

Found: '12' is likely a % command; tubes are unsteady at low power[8].

Impact: every strip size is provisional.

How: LaserCAD config audit.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



Also Caught: STEP exported in cm (now mm); a drawing showed 0.13 for 0.125 mm (fixed).

Potential Solutions & Risk Mitigation

FIXED · DESIGNED · FUTURE

Fixed & Verified

Seal: starts at x = 23 mm; 0 clash.

Units: all STEP files in mm.

Checks: 99 CAD checks pass.

Designed, to Prove

v0.2 strip: every sense lead looped.

Contact: silver pads, rounded pins.

Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.

Process: slower passes, log current.

Concept: one-zone strip (score 4.29).

Future Scope & Validation Plan

- **Experiments:** E2 laser, reader bench test, E4 wet tests.
- **Material:** Raman and SEM before any 'graphene' claim.
- **Reliability:** bending, contact cycling, humidity drift.
- **Product:** battery pod; phone app on the BLE broadcast; strips per dressing.
- **Before patients:** ISO 10993, IEC 60601-1, ethics approval.
- **Not claimed:** sensing performance, graphene, clinical benefit.

Rule We Keep: no performance, graphene or clinical claim until it is measured.

References & Sources

SOURCES WITH READING DEPTH

Numbers match the superscripts on each slide. After each source we state how deeply it was read; nothing is labelled verified beyond that.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read

2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts

3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract

4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read

5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage

6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)

7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read

8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed

9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract

10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network

11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page

12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents

13. Wounds UK news (22 Aug 2018) on a Mölnlycke-commissioned survey of >200 UK patients with chronic wounds: average dressing time 18 min · patient-reported, industry-sponsored; web page read

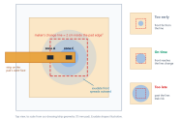
14. Smith+Nephew, ALLEVYN LIFE product page: visual change indicator ('At 75%, it's time to change your dressing') · manufacturer web page, read 19 Oct 2026

15. Truemeds.in retail listings: Mepilex Lite 10 × 10 cm ₹242; Mepilex Border 10 × 10 cm ₹514 (pack size unclear) · retail pages, read 19 Oct 2026

Image Sources

ALL IMAGES FROM OUR FILES

Every image comes from our own project files: SolidWorks renders and screenshots, the KiCad export, and plots drawn by our scripts (written with AI assistance). No stock photos and no AI image generators. Team photos (slides 2 and 12) are taken by the team.



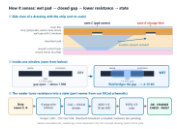
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
1_presentation/figures/fig_problem.png



Before-state pain-point map
Source: make_figures.py · cited sources 1–3
1_presentation/figures/fig_journey_before.png



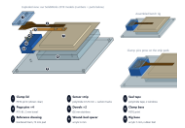
After-state intended workflow
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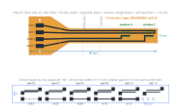
How it senses (concept)
Source: make_figures.py · sensor_selection.md, calc_readout.py
1_presentation/figures/fig_howitworks.png



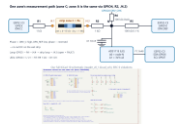
Firmware scan and state logic
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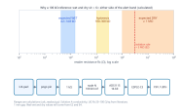
Annotated SolidWorks views
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1_presentation/figures/fig_cad.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
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Measurement path + KiCad sheet
Source: make_figures.py · reader_v0_1 netlist + KiCad PDF
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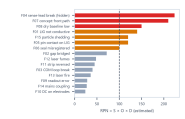
Measurement window
Source: make_figures.py · calc_readout.py
1_presentation/figures/fig_window.png



Evidence gallery + tool output
Source: our files, screen capture, 19 Oct tool runs
1_presentation/figures/fig_maker.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
1_presentation/figures/fig_roadmap.png



Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png